# Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

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# Dataset at a Glance

**3,900**

Purchases

**18**

Features

**$59.76**

Avg. Spend

**3.75**

Avg. Rating

Key features: demographics, purchase details, shipping, discounts, review ratings, and subscription status.

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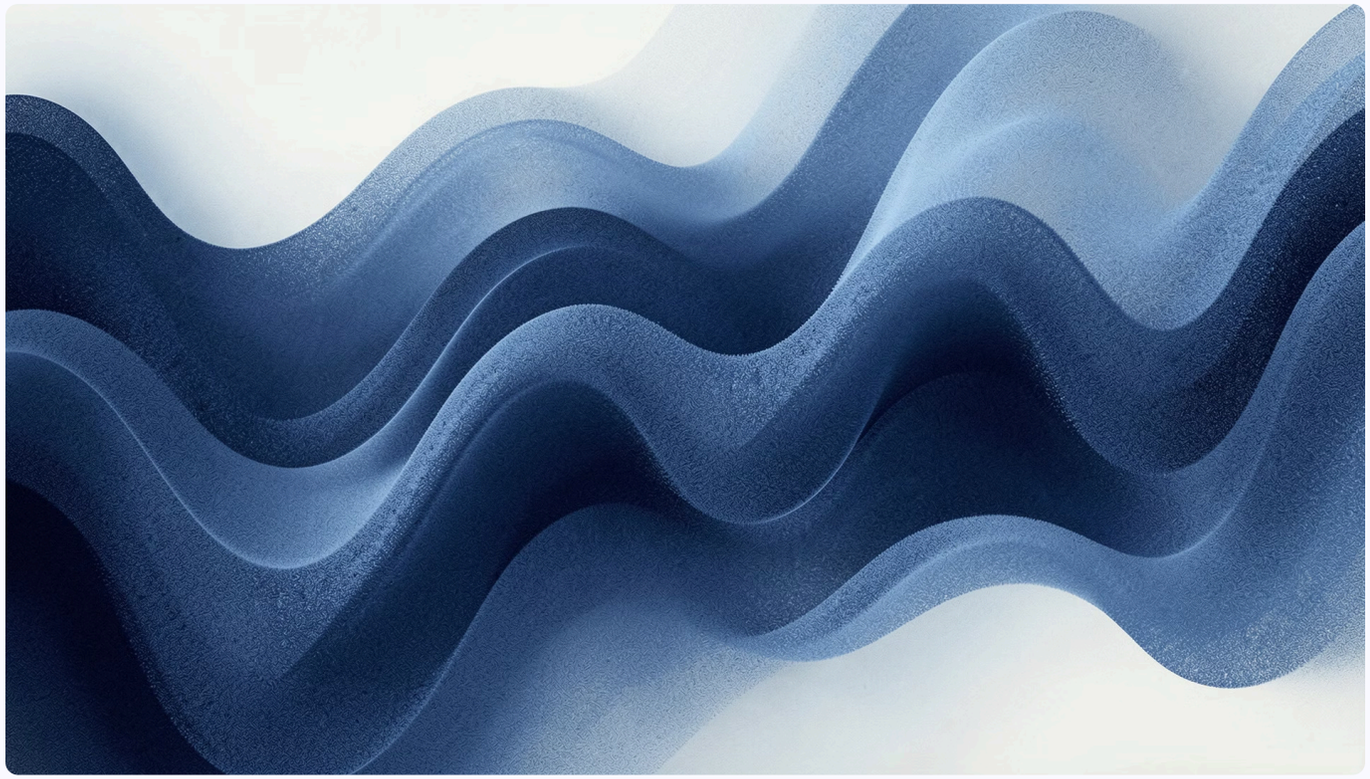
# Data Preparation in Python



01

## Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

## Clean & Engineer

Imputed missing ratings; created `age_group` and `purchase_frequency_days`

03

## Standardize & Load

Snake case columns; dropped redundant `promo_code_used`; loaded to PostgreSQL

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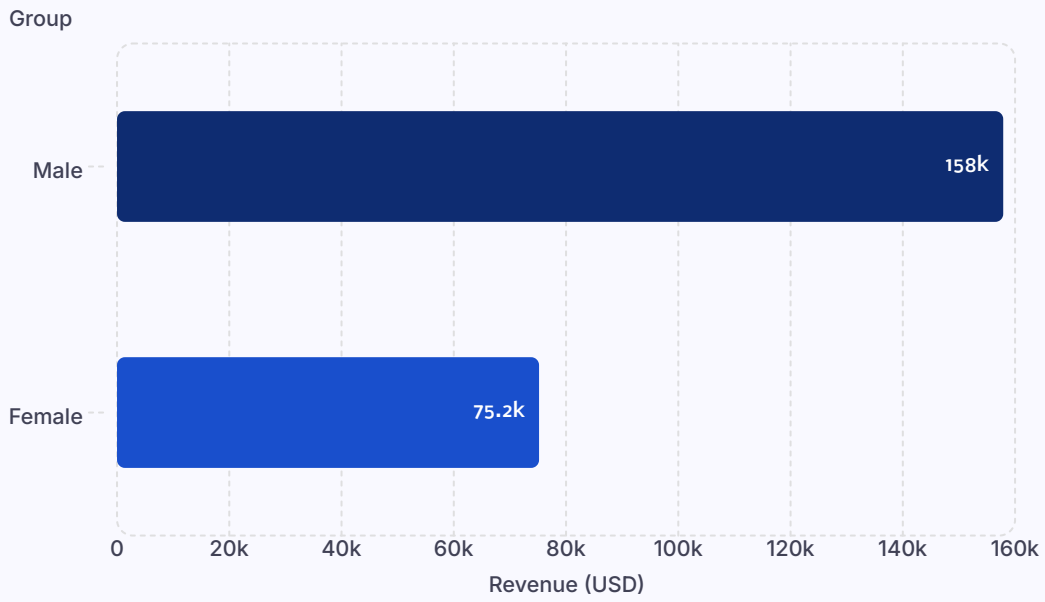
# Revenue & Spending Insights




## Revenue by Gender

Male customers generate **2× more revenue** than female customers.

## Shipping & Spend

Express shipping users spend slightly more (**$60.48**) vs. Standard (**$58.46**).

## High-Spend Discount Users

**839 customers** used discounts yet spent above average – a key retention segment.

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# Top-Rated Products

| | |
|---|---|
| 1 | Gloves<br>Avg. rating **3.86** |
| 2 | Sandals<br>Avg. rating **3.84** |
| 3 | Boots<br>Avg. rating **3.82** |
| 4 | Hat<br>Avg. rating **3.80** |
| 5 | Skirt<br>Avg. rating **3.78** |

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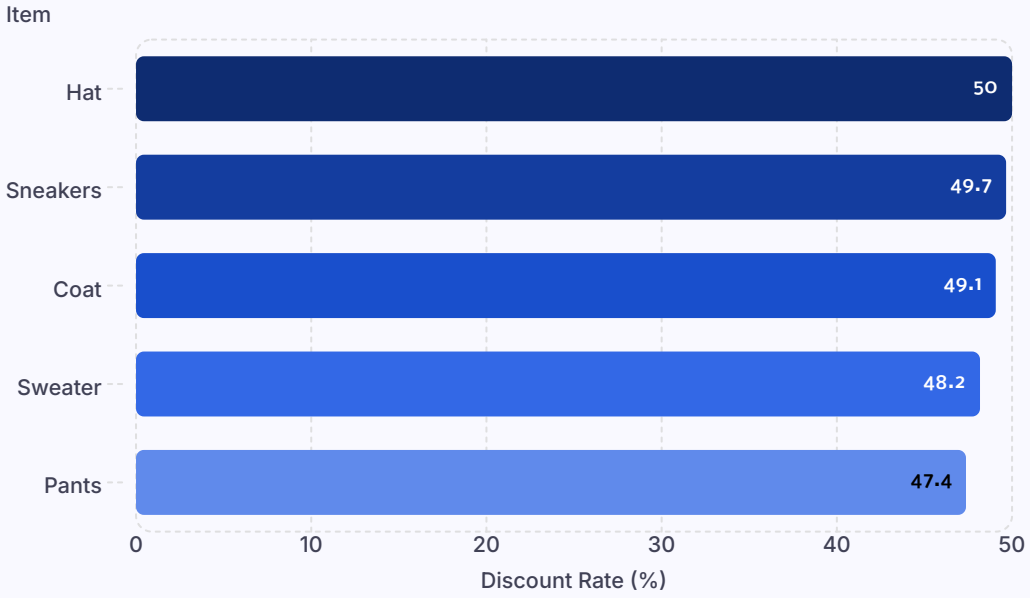
# Discount-Dependent Products




## Top 5 Discount-Reliant Items

Nearly **50% of Hat purchases** used a discount – signaling potential margin risk.

All five items exceed **47% discount rate**, warranting a pricing policy review.

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# Customer Segmentation

### Loyal

**3,116** customers

### Returning

**701** customers

### New

**83** customers

Repeat buyers (>5 purchases): **958 subscribers** vs. **2,518 non-subscribers** – loyalty programs can close this gap.

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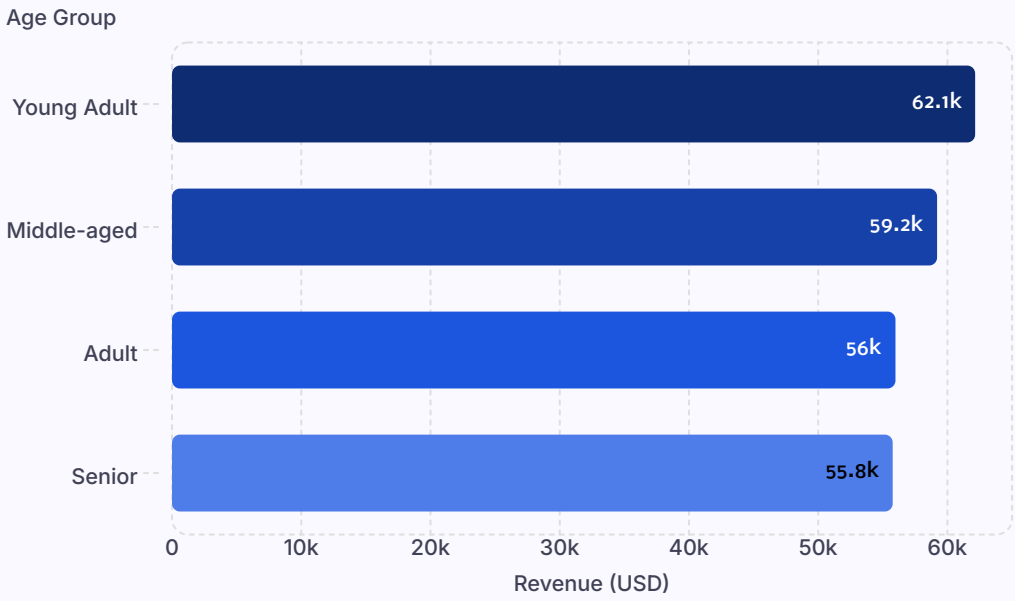
# Revenue by Age Group & Subscriptions




## Subscribers vs. Non-Subscribers

| Status | Customers | Avg. Spend | Total Revenue |
|---|---|---|---|
| Subscribed | 1,053 | $59.49 | $62,645 |
| Non-subscribed | 2,847 | $59.87 | $170,436 |

**Young Adults** lead revenue – a prime target for subscription campaigns.

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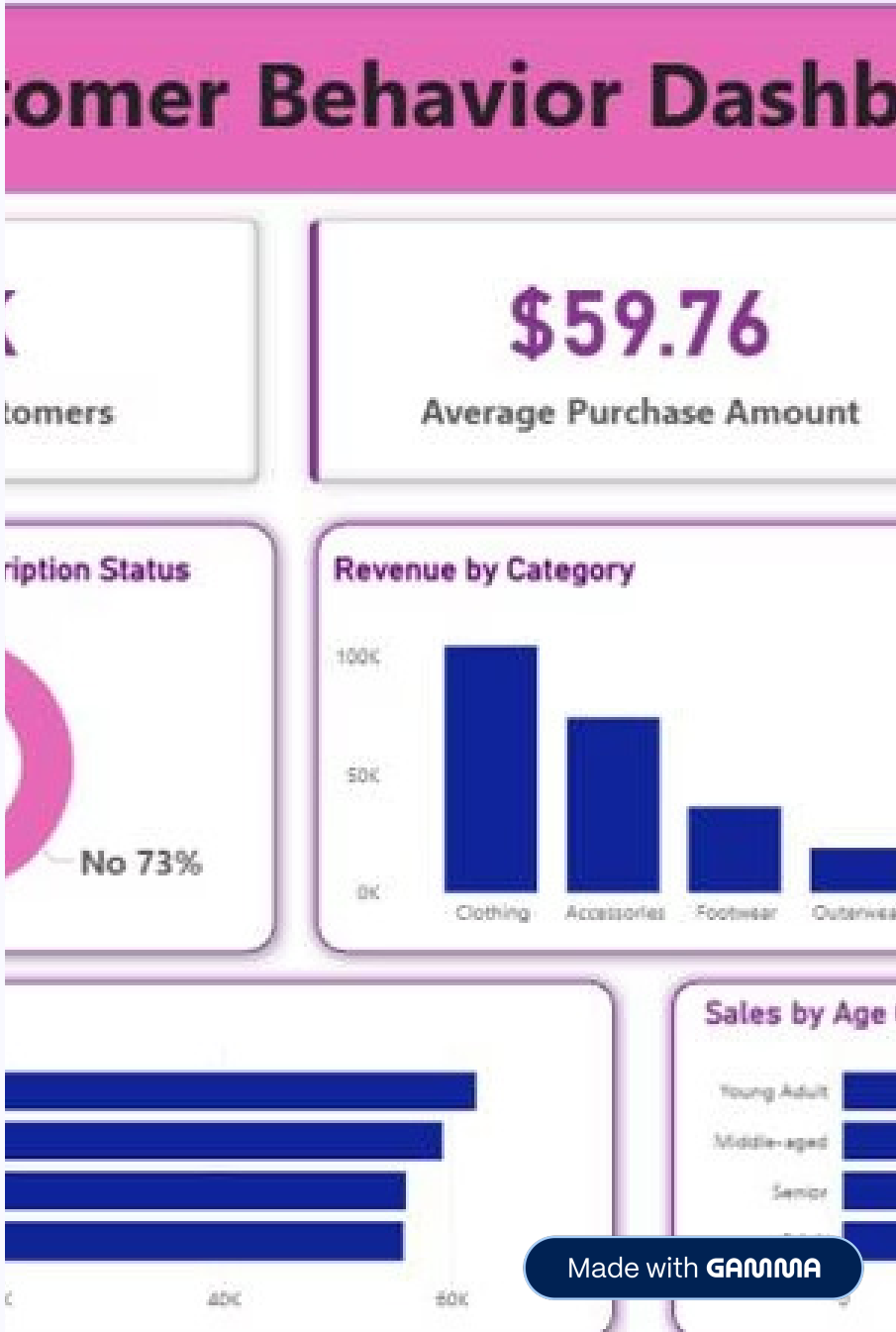
# Power BI Dashboard

Interactive dashboard with filters for gender, category, subscription, and shipping – surfacing KPIs and category/age-group breakdowns at a glance.

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# Business Recommendations

### Boost Subscriptions

Promote exclusive subscriber benefits

### Loyalty Programs

Reward repeat buyers to grow the Loyal segment

### Review Discount Policy

Balance sales boosts with margin control

### Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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